Air-Quality Prediction for Nashik Monitoring Stations Using KNN Imputation and a CNN-LSTM Network

Academic Project Report

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[Course / guide]

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Abstract

Hourly concentrations of seven pollutants (PM2.5, PM10, NO2, SO2, CO, O3 and NH3) from four continuous monitoring stations in Nashik, India (MIDC Ambad, Gangapur Road, Pandav Nagari and Hirwadi) were used to build a one-hour-ahead prediction model. After rule-based cleaning, missing inputs were filled with a K-nearest-neighbours imputer fitted on training-period data only. A separate convolutional-LSTM (CNN-LSTM) network was trained for each station on 24-hour input windows, with the latest calendar year of each station held out for testing: January-August 2026 for three stations (July 2026 is absent from the files) and 2025 for Pandav Nagari, whose record ends on 31 December 2025. Predictions were scored in original units with MSE, RMSE, MAE and R^2 against observed values only, and compared with a persistence baseline on identical test hours. Pooled over stations, with four flagged sensor series excluded, the CNN-LSTM reached test R^2 between 0.46 (NH3) and 0.85 (PM10), and reduced RMSE relative to persistence by: PM2.5 5.3%, PM10 8.6%, NO2 9.9%, SO2 8.3%, CO 10.6%, O3 9.7%, NH3 3.7%. A network trained to predict concentration levels directly was worse than persistence on validation data at every station; predicting the change from the last observed value was therefore selected on validation loss. The gains over persistence are consistent but modest, and four station-pollutant test series show sensor-quality problems that dominate their scores.

Keywords

air-quality prediction; CNN-LSTM; KNN imputation; time-series forecasting; persistence baseline; Nashik; particulate matter; chronological validation

1. Introduction

Short-term air-quality prediction supports public advisories and operational decisions such as traffic or industrial-emission controls. Continuous ambient monitoring stations report hourly concentrations, but their records contain gaps, instrument faults and changes in data availability, so any prediction pipeline must handle missing data without leaking information from the future.

Hybrid convolutional-recurrent networks are a common choice for such series: one-dimensional convolutions extract short local patterns, and Long Short-Term Memory (LSTM) layers model longer dependencies [1], [3], [4]. This project applies that design to data from four Nashik stations and evaluates it against the simplest competitive forecast, persistence.

2. Problem Statement

Given the hourly pollutant observations of a Nashik monitoring station up to hour t, predict the concentrations of seven pollutants at hour t+1, and quantify the accuracy of those predictions over the latest calendar year in the data, which is excluded from model fitting and model selection.

3. Objectives

- Inspect the Nashik dataset and document its schema, coverage and quality problems.
- Build a reproducible cleaning pipeline and KNN imputation fitted without temporal leakage.
- Design, train and validate a CNN-LSTM model for next-hour prediction at each station.
- Evaluate predictions with MSE, RMSE, MAE and R^2 in original units per station and pollutant, and compare them with a persistence baseline on identical test hours.
- Report limitations, including data-quality issues that affect the evaluation.

4. Dataset Description

The dataset consists of four CSV files, one per station. Station identity is given only by the file name; no column identifies the station. All files share the same 25 columns: a Timestamp plus 24 measurements (criteria pollutants, volatile organic compounds and meteorological variables). The Hirwadi file has a 26th column, Source_File, whose labels do not match the timestamps (rows labelled '2023.csv.csv' carry 2026 timestamps), so it was dropped and the timestamps were treated as authoritative. Hirwadi rows are also stored out of chronological order and were sorted. The column naming matches the export format of India's Central Pollution Control Board continuous-monitoring portal; the provenance should be confirmed with the data provider before publication.

Table 1. Raw files as delivered.

Station	Rows	First timestamp	Last timestamp	Duplicate times	Sorted	Step	Years present	Rows with no pollutant
MIDC Ambad	40152	2017-01-01	2026-08-31 23:00:00	0	True	1 hour	2017, 2023, 2024, 2025, 2026	14377
Gangapur Road	81024	2017-01-01	2026-08-31 23:00:00	0	True	1 hour	2017, 2018, 2019, 2020, 2021, 2022,	4945

							2023, 2024, 2025, 2026	
Pandav Nagari	35064	2017-01-01	2025-12-31 23:00:00	0	True	1 hour	2017, 2023, 2024, 2025	12761
Hirwadi	40152	2017-01-01	2026-08-31 23:00:00	0	False	1 hour	2017, 2023, 2024, 2025, 2026	13873

One observation is one hourly average. Three files begin with a full year of empty 2017 placeholder rows and resume in mid-2023 (the first observation is on 14 June 2023 at MIDC Ambad, 22 May 2023 at Pandav Nagari and 23 May 2023 at Hirwadi). Gangapur Road has the longest record, starting in 2017 with a 123-day gap in that year. In all files that reach 2026, July 2026 is missing entirely. The latest calendar year is therefore 2026, covering 1 January-30 June and 1-31 August (5,088 of 8,760 hours); it is incomplete. Pandav Nagari has no 2026 rows; its latest year, 2025, is complete.

Seven pollutants were selected as prediction targets: the pollutants of the Indian National AQI that are present in the data (PM2.5, PM10, NO2, SO2, CO, O3, NH3). The aromatic VOCs (benzene, toluene, xylenes) were not used. Xylene, BP and VWS are empty in every file. Meteorological variables were not used as inputs: air temperature and relative humidity are absent at Gangapur Road before 2026, and wind speed shows level shifts that look like sensor changes (for example, a median of 0.0 m/s at MIDC Ambad in 2026 against about 0.5 m/s before, and isolated values above 20 m/s at Gangapur Road and Hirwadi). The full data dictionary, with per-station missing percentages, is exported as outputs/tables/data_dictionary.csv.

Table 2. Data dictionary for the columns used by the model.

Column (standardised)	Original column	Unit	Role
Timestamp	Timestamp	-	Time index (hourly)
PM2.5	PM2.5 (µg/m³)	µg/m³	Model input and prediction target
PM10	PM10 (µg/m³)	µg/m³	Model input and prediction target
NO2	NO2 (µg/m³)	µg/m³	Model input and prediction target
SO2	SO2 (µg/m³)	µg/m³	Model input and prediction target
CO	CO (mg/m³)	mg/m³	Model input and prediction target
O3	Ozone (µg/m³)	µg/m³	Model input and prediction target
NH3	NH3 (µg/m³)	µg/m³	Model input and prediction target

5. Data Preprocessing and KNN Imputation

5.1 Cleaning rules

- Timestamps parsed and sorted; duplicate timestamps would keep the first row (none were found).
- Rows before a station's first pollutant observation removed (the empty 2017 placeholder blocks): 12,710 rows at MIDC Ambad, 12,161 at Pandav Nagari and 12,184 at Hirwadi.

- Negative concentrations set to missing (none occurred).
- PM2.5 or PM10 values of 999 or more set to missing, because that is the analyser ceiling, not a measurement (21 values, all at Gangapur Road).
- Any value repeated unchanged for 12 or more consecutive hours set to missing as a stuck-sensor reading. This removed 13,921 values, mostly CO readings of exactly 0.00 mg/m^3 and long flat NO2 stretches at Gangapur Road.
- No interpolation, aggregation or resampling. Hours absent from the files (the 2017 gap at Gangapur Road and July 2026) were not filled, and no input window spans them.

Table 3. Values removed by the stuck-sensor rule (12 or more identical consecutive hours).

Station	PM2.5	PM10	NO2	SO2	CO	O3	NH3
MIDC Ambad	22	0	0	0	3059	14	12
Gangapur Road	346	124	2853	193	4005	155	806
Pandav Nagari	0	0	0	0	0	0	0
Hirwadi	13	13	59	0	2247	0	0

5.2 Missing values

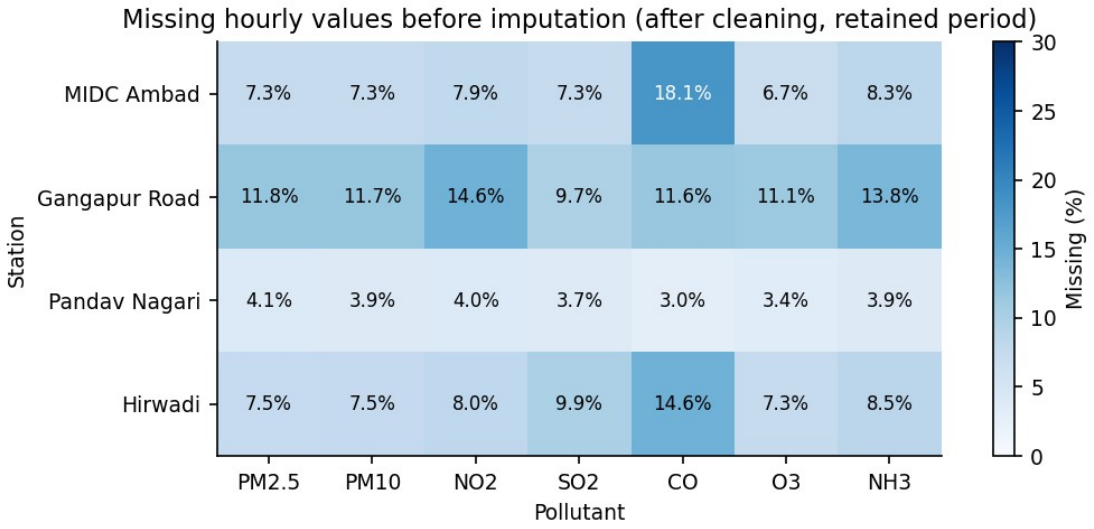


Figure 1. Missing hourly values per station and pollutant before imputation, after cleaning, over each station's retained period.

5.3 KNN imputation

K-nearest-neighbours imputation fills a missing value with a weighted average of that variable in the k most similar records [2], [5]. Similarity is measured by the NaN-Euclidean distance, which compares two hours over the variables both have observed and rescales for the missing ones. Here k = 5 and neighbours are weighted by inverse distance (scikit-learn KNNImputer [6]).

- Features: the seven standardised pollutant concentrations of the same hour. Station identifiers and timestamps are not used in the distance.

- One StandardScaler and one KNNImputer per station, fitted on that station's training rows only and then applied to validation and test rows. Neighbours (donors) are therefore always training-period hours. Scaling is needed because units and ranges differ widely, for example CO in mg/m^3 against PM10 in hundreds of µg/m^3.
- Each station is imputed from its own history, preserving station-specific pollutant relationships. Stations never donate to each other.
- Imputed values are used only as model inputs. A missing value at hour s is filled from other pollutants observed at hour s, which are known when a prediction for any later hour is issued. Prediction targets are never imputed: training loss and every evaluation metric use observed values only.
- Hours with every pollutant missing receive the training mean (no distance can be computed). Input windows in which more than half of the pollutant cells are imputed are discarded (Section 9).

Table 4. Missing pollutant cells before and after KNN imputation (7 pollutants).

Station	Split	Missing cells before	Missing cells after
MIDC Ambad	train	9802	0
MIDC Ambad	val	1664	0
MIDC Ambad	test	5818	0
Gangapur Road	train	58622	0
Gangapur Road	val	1423	0
Gangapur Road	test	8310	0
Pandav Nagari	train	1898	0
Pandav Nagari	val	1121	0
Pandav Nagari	test	2936	0
Hirwadi	train	11641	0
Hirwadi	val	1014	0
Hirwadi	test	5052	0

6. Exploratory Data Analysis

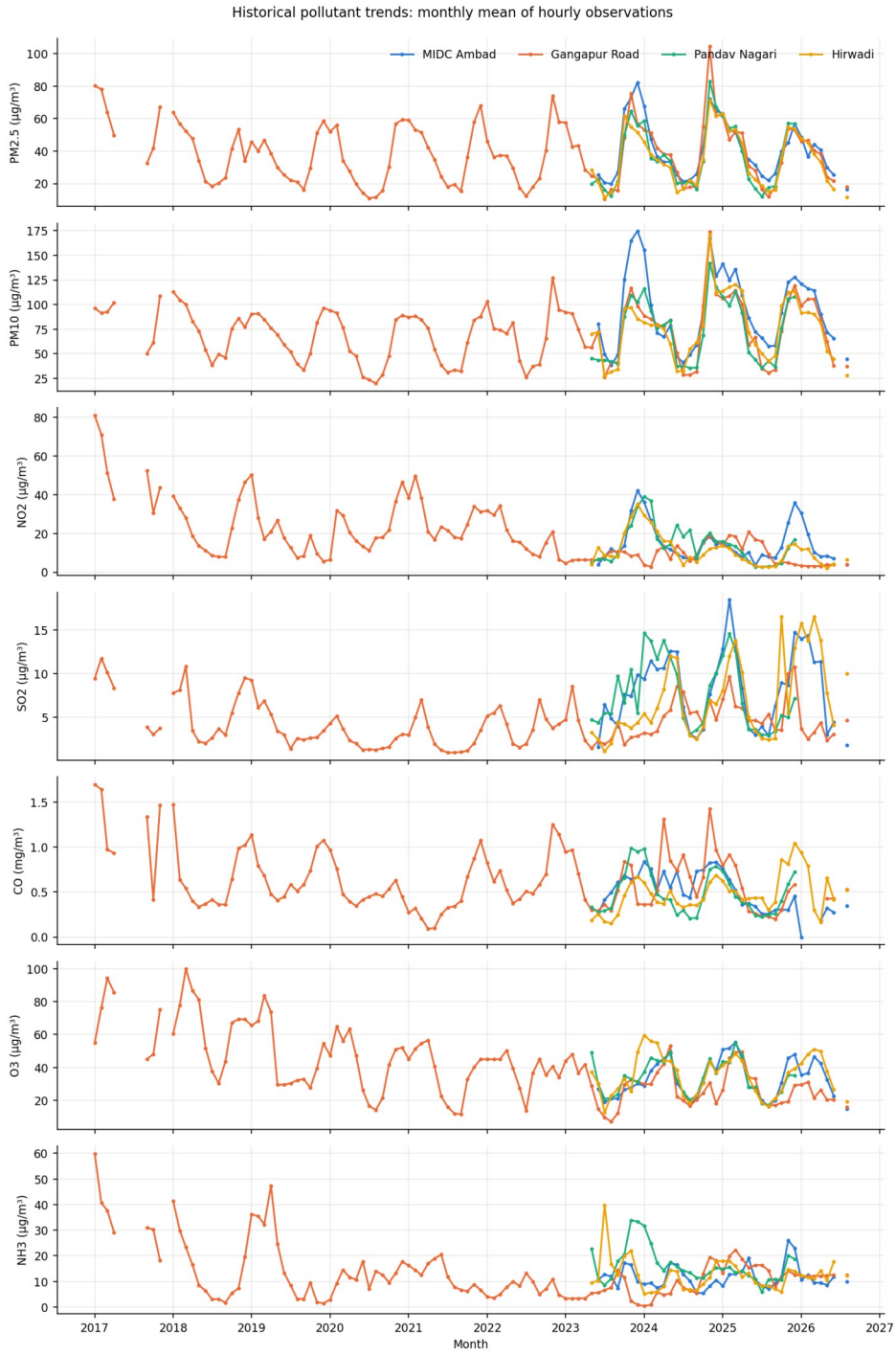


Figure 2. Monthly means of hourly observations per pollutant and station. Gaps are periods with no observations.

Particulate matter shows a seasonal cycle at all stations, with lower concentrations in the monsoon months (June-September) and higher concentrations in winter. Gangapur Road is the only station with history before 2023. Its NO2, CO and NH3 series change character in 2026: NO2 settles near 3 µg/m³ in

0.25 steps and CO near 0 mg/m³, consistent with an instrument or calibration change rather than a real drop in pollution.

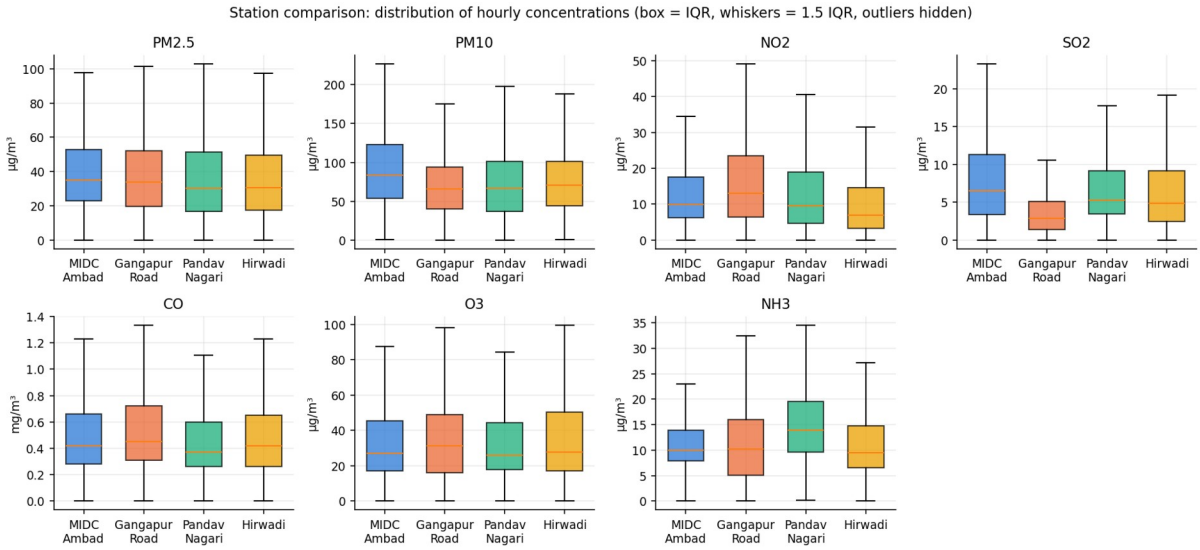


Figure 3. Distribution of hourly concentrations by station (box: interquartile range; whiskers: 1.5 IQR; outliers hidden).

Table 5. Percentage of missing hourly values before imputation (retained period).

Station	PM2.5	PM10	NO2	SO2	CO	O3	NH3
MIDC Ambad	7.3	7.3	7.9	7.3	18.1	6.8	8.3
Gangapur Road	11.8	11.7	14.6	9.7	11.6	11.2	13.8
Pandav Nagari	4.1	3.9	4.0	3.7	3.0	3.4	3.9
Hirwadi	7.5	7.5	8.0	9.9	14.6	7.3	8.5

7. Proposed CNN-LSTM Architecture

A one-dimensional convolution slides a small learned filter along the time axis. With kernel size 3, each output step combines three consecutive hours of all input features, so a filter can respond to a local motif such as a sharp morning rise in NO2 [7]. Stacking two such layers widens the receptive field to five hours. Causal padding ensures that no output step depends on later steps. Max-pooling then halves the sequence, keeping the strongest response in each pair of steps.

An LSTM layer [1] reads the pooled sequence step by step. Its cell state is updated through input, forget and output gates, which lets it carry information across the whole window without the vanishing gradients of a plain recurrent network. The final hidden state summarises the 24-hour history; dropout [9] regularises it, and two dense layers map it to the seven outputs.

Table 6. CNN-LSTM layers (50,247 trainable parameters per station model).

Layer	Output shape	Configuration	Parameters
Input	(24, 13)	24 hours x 13 features	0
Conv1D	(24, 64)	64 filters, kernel 3, causal	2,560

		padding, ReLU	
Conv1D	(24, 64)	64 filters, kernel 3, causal padding, ReLU	12,352
MaxPooling1D	(12, 64)	pool size 2	0
LSTM	(64)	64 units, final hidden state	33,024
Dropout	(64)	rate 0.2	0
Dense	(32)	ReLU	2,080
Dense (output)	(7)	linear, one unit per pollutant	231

Inputs per hour: the seven standardised, KNN-imputed pollutant concentrations plus sine and cosine encodings of hour-of-day, day-of-week and month (13 features). Calendar encodings are deterministic and known in advance. The model is multi-output: one network predicts all seven pollutants of a station, and each pollutant is evaluated separately.

8. Methodology and Experimental Setup

8.1 Forecasting design

The task is one-step-ahead (next-hour) prediction. To predict hour t+1, the model receives the 24 hours t-23 … t of observed (and, where missing, imputed) inputs. In the test year the inputs are the real observations preceding each predicted hour, exactly as they would be available to an operational system that updates every hour. No prediction is fed back as an input, so this is not a recursive forecast. The test year is a held-out historical year with known measurements. The work does not forecast genuinely future hours, for which no measurements exist, and its accuracy says nothing about predicting more than one hour ahead.

8.2 Station-specific models

Each station has its own model, scaler and imputer. A shared model was not used because the stations' histories differ sharply (Gangapur Road from 2017; the others from mid-2023) and their test years differ (Pandav Nagari ends in 2025). In a shared model, Pandav Nagari's 2025 test year would coincide with training data from the other stations, and correlated regional weather would leak across stations. Independent models avoid that and keep sequences from crossing station boundaries by construction.

8.3 Sequence construction

- Lookback 24 hours (one diurnal cycle); horizon 1 hour.
- A window is kept only if its 25 timestamps are consecutive hours, so no window spans a gap in the file.
- At least one of the seven targets must be observed at the predicted hour.
- At most 50% of the window's input pollutant cells may be imputed.
- Every pollutant must have an earlier observation (needed by the baseline).

8.4 Training configuration and target design

Table 7. Training configuration (identical for all stations; no hyperparameter search).

Setting	Value

Loss	Masked MSE on standardised targets (missing targets excluded)
Optimiser	Adam [8], learning rate 0.001
Batch size / maximum epochs	256 / 60
Early stopping	validation loss, patience 6, best weights restored
Random seed	42 (Python, NumPy, TensorFlow)
Software	Python 3.12.14, TensorFlow 2.19.0, scikit-learn 1.6.1, pandas 2.2.2, NumPy 2.0.2

Two parameterisations of the same network were trained per station, and the one with the lower validation loss was selected. In the level design, the network predicts the standardised concentration at t+1. In the change design, it predicts the difference between the concentration at t+1 and the last observed value, which is then added back. Because both are scored in the same standardised units, their validation losses are directly comparable with each other and with persistence.

Table 8. Model selection on the validation period (masked MSE, standardised units). The test year played no part.

Station	Train seq.	Val seq.	Val loss: persistence	Val loss: level	Val loss: change	Best epoch (change)	Selected
MIDC Ambad	16498	4178	0.1625	0.1817	0.1387	13	change
Gangapur Road	64835	4223	0.1109	0.1293	0.09261	9	change
Pandav Nagari	9499	4264	0.216	0.2312	0.1756	16	change
Hirwadi	16951	4303	0.3727	0.412	0.3065	5	change

The level design was worse than persistence on validation at all four stations. The change design beat persistence at all four, so it was selected everywhere. For transparency: the first version of the pipeline used only the level design. Its test-year results were computed before the change design was added, and they were worse than persistence. The switch is justified by the validation evidence above, which was available without the test year. Test RMSE for both designs is kept in outputs/metrics/station_pollutant_metrics.csv.

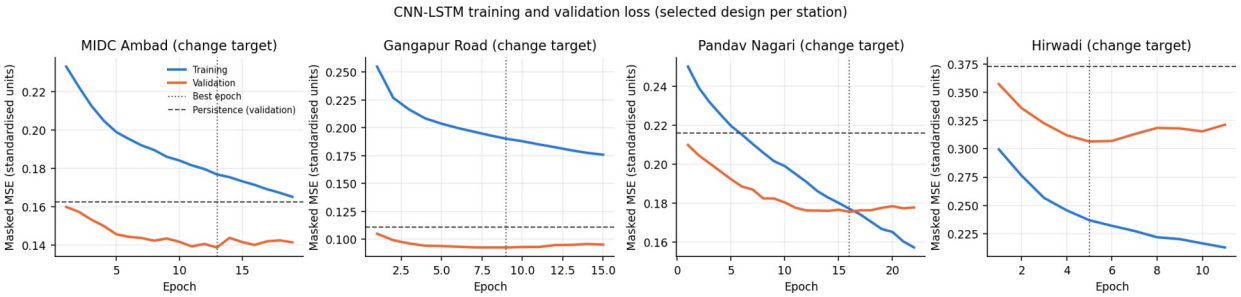


Figure 4. Training and validation loss of the selected (change) design per station. The dashed line is persistence on the validation period.

Validation loss is below training loss at three stations. This is expected rather than suspicious: dropout is active only during training, and the validation half-year (July-December) includes the monsoon months, when concentrations are low and smooth. Hirwadi is the exception: validation loss is higher and rises after epoch 5, showing early overfitting, which early stopping caught.

9. Train-Validation-Test Strategy

Random splitting would put hours from the same day into both training and test sets. Because neighbouring hours are highly correlated, the model would be scored on near-copies of its training data, and it would also learn from the future of the period it is tested on [10]. A chronological split avoids both problems. For each station, the test set is its latest calendar year, validation is the six months before it, and training is everything earlier. Windows are assigned to a split by the timestamp of the predicted hour. Scaler and imputer statistics come from training rows only.

Table 9. Chronological splits per station.

Station	Split	Start	End	Hourly rows	Months	PM2.5 observed
MIDC Ambad	train	2023-06-14 14:00:00	2025-06-30 23:00:00	17938	25	16560
MIDC Ambad	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	6	4175
MIDC Ambad	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	7	4708
Gangapur Road	train	2017-01-01 00:00:00	2025-06-30 23:00:00	71520	98	62415
Gangapur Road	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	6	4197
Gangapur Road	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	7	4876
Pandav Nagari	train	2023-05-22 17:00:00	2024-06-30 23:00:00	9727	14	9439
Pandav Nagari	val	2024-07-01 00:00:00	2024-12-31 23:00:00	4416	6	4251
Pandav Nagari	test	2025-01-01 00:00:00	2025-12-31 23:00:00	8760	12	8277
Hirwadi	train	2023-05-23 16:00:00	2025-06-30 23:00:00	18464	26	16799
Hirwadi	val	2025-07-01 00:00:00	2025-12-31 23:00:00	4416	6	4306
Hirwadi	test	2026-01-01 00:00:00	2026-08-31 23:00:00	5088	7	4769

10. Evaluation Metrics: MSE, RMSE, MAE and R^2

For n test hours with observed values y_i, predictions $\hat{y}_i$ and observed mean $\bar{y}$:

$$MSE = (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE}$$

$$MAE = (1/n)\ \Sigma\ |y_i - \hat{y}_i|$$

$$R^2 = 1 - \Sigma\ (y_i - \hat{y}_i)^2\ /\ \Sigma\ (y_i - \bar{y})^2$$

- MSE is in squared units (for example $(\mu g/m^3)^2$). It penalises large errors heavily, so pollution spikes dominate it.
- RMSE is in the pollutant's own unit and is the typical size of an error, weighted towards large errors.
- MAE is in the pollutant's unit and is the average absolute error. It is less sensitive to spikes than RMSE.

- R^2 is unitless: the share of the test-period variance explained by the predictions. 1 is perfect; 0 equals always predicting the test mean. It is negative when predictions are worse than that, which is possible on test data, and undefined when the test values are constant or there are fewer than two of them (reported as missing in that case).
- RMSE skill = 1 − RMSE_model / RMSE_persistence. Positive values mean the CNN-LSTM beats persistence; 0.05 means a 5% lower RMSE.

All metrics are computed with scikit-learn in original units, after inverse transformation and only on hours with an observed value. Pollutants are never pooled into one score because their units and scales differ. The overall summary is computed per pollutant by concatenating the test hours of all stations whose series is not quality-flagged.

11. Experimental Results

*Table 10. CNN-LSTM test results per station and pollutant, in original units (µg/m³; CO in mg/m³, MSE in squared units). * = quality-flagged test series (fewer than 100 distinct values; see Section 14).*

Station	Pollutant	Evaluation Period	Number of Test Observations	MSE	RMSE	MAE	R^2
MIDC Ambad	PM2.5	2026-01-01 to 2026-08-31	4542	35.71	5.976	3.981	0.8656
MIDC Ambad	PM10	2026-01-01 to 2026-08-31	4544	238.6	15.45	8.786	0.8544
MIDC Ambad	NO2	2026-01-01 to 2026-08-31	4467	25.87	5.086	2.794	0.8402
MIDC Ambad	SO2	2026-01-01 to 2026-08-31	4542	5.725	2.393	1.278	0.8637
MIDC Ambad	CO *	2026-04-21 to 2026-08-31	1574	0.007139	0.08449	0.05104	0.4433
MIDC Ambad	O3	2026-01-01 to 2026-08-31	4552	41.55	6.446	4.464	0.8903
MIDC Ambad	NH3	2026-01-01 to 2026-08-31	4489	1.908	1.381	0.8995	0.7476
Gangapur Road	PM2.5	2026-01-01 to 2026-08-31	4800	135.4	11.64	5.337	0.7073
Gangapur Road	PM10	2026-01-01 to 2026-08-31	4780	350.1	18.71	10.8	0.8108
Gangapur Road	NO2 *	2026-01-02 to 2026-08-31	2190	1.426	1.194	0.4078	-1.101
Gangapur Road	SO2	2026-01-01 to 2026-08-31	4259	7.392	2.719	1.263	0.2832
Gangapur Road	CO *	2026-05-15 to 2026-08-31	1705	0.003825	0.06184	0.05053	-0.5708
Gangapur Road	O3	2026-01-01 to 2026-08-31	4905	53.51	7.315	3.199	0.687
Gangapur Road	NH3 *	2026-01-01 to 2026-08-31	4198	3.911	1.978	0.3034	-0.8288
Pandav Nagari	PM2.5	2025-01-02 to 2025-12-31	8135	93.15	9.652	3.762	0.8659
Pandav Nagari	PM10	2025-01-02 to 2025-12-31	8161	220.7	14.86	7.161	0.8919
Pandav Nagari	NO2	2025-01-02 to 2025-12-31	8199	15.57	3.945	2.081	0.7819
Pandav Nagari	SO2	2025-01-02 to 2025-12-31	8177	34.22	5.85	1.999	0.6176
Pandav Nagari	CO	2025-01-02 to 2025-12-31	8266	0.01912	0.1383	0.07446	0.8104
Pandav Nagari	O3	2025-01-02 to 2025-12-31	8202	49.97	7.069	4.604	0.8857
Pandav Nagari	NH3	2025-01-02 to 2025-12-31	8216	22.02	4.692	2.044	0.6167

Hirwadi	PM2.5	2026-01-01 to 2026-08-31	4632	20.22	4.497	3.079	0.9311
Hirwadi	PM10	2026-01-01 to 2026-08-31	4628	305.6	17.48	7.233	0.765
Hirwadi	NO2	2026-01-01 to 2026-08-31	4402	11.66	3.414	2.091	0.7002
Hirwadi	SO2	2026-01-01 to 2026-08-31	4436	64.96	8.06	2.881	0.5709
Hirwadi	CO	2026-01-02 to 2026-08-31	2407	0.03036	0.1742	0.08154	0.773
Hirwadi	O3	2026-01-01 to 2026-08-31	4642	159.5	12.63	6.087	0.7411
Hirwadi	NH3	2026-01-01 to 2026-08-31	4508	64.75	8.047	3.561	0.2009

Table 11. Overall summary per pollutant: test hours of all unflagged stations pooled. Test periods: 2025 (Pandav Nagari) and January-August 2026 (others).

Pollutant	Unit	Test obs.	MSE	RMSE	MAE	R^2	Stations included
PM2.5	µg/m³	22109	75.24	8.674	4.006	0.8423	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
PM10	µg/m³	22113	270.1	16.44	8.296	0.8512	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NO2	µg/m³	17068	17.25	4.154	2.27	0.8123	MIDC Ambad, Pandav Nagari, Hirwadi
SO2	µg/m³	21414	29.21	5.404	1.883	0.6538	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
CO	mg/m³	10673	0.02165	0.1471	0.07605	0.8062	Pandav Nagari, Hirwadi
O3	µg/m³	22301	71.83	8.475	4.575	0.8339	MIDC Ambad, Gangapur Road, Pandav Nagari, Hirwadi
NH3	µg/m³	17213	27.97	5.288	2.143	0.4613	MIDC Ambad, Pandav Nagari, Hirwadi

11.1 Station-wise interpretation

MIDC Ambad. Test period 2026-01-01 to 2026-08-31 (2026), 1,574-4,552 observed hours per pollutant. R^2 ranges from 0.44 (CO) to 0.89 (O3). The model beats persistence on RMSE for 5 of 6 unflagged pollutants, with a mean RMSE skill of 8.5%. Flagged series: CO (skill -0%).

Gangapur Road. Test period 2026-01-01 to 2026-08-31 (2026), 1,705-4,905 observed hours per pollutant. R^2 ranges from -1.10 (NO2) to 0.81 (PM10). The model beats persistence on RMSE for 4 of 4 unflagged pollutants, with a mean RMSE skill of 4.6%. Flagged series: NO2, CO, NH3 (skill -9%, -420%, 3%).

Pandav Nagari. Test period 2025-01-02 to 2025-12-31 (2025), 8,135-8,266 observed hours per pollutant. R^2 ranges from 0.62 (NH3) to 0.89 (PM10). The model beats persistence on RMSE for 7 of 7 unflagged pollutants, with a mean RMSE skill of 9.3%.

Hirwadi. Test period 2026-01-01 to 2026-08-31 (2026), 2,407-4,642 observed hours per pollutant. R^2 ranges from 0.20 (NH3) to 0.93 (PM2.5). The model beats persistence on RMSE for 7 of 7 unflagged pollutants, with a mean RMSE skill of 5.3%.

Pandav Nagari and MIDC Ambad gain most over persistence (mean skill 9.3% and 8.5%), with the largest gains for O3 at both stations and for NO2 and PM10 at MIDC Ambad; these pollutants have strong diurnal cycles that a 24-hour window captures. Gangapur Road has the most training data but the weakest test behaviour, because three of its seven test series changed character in 2026. Hirwadi's NH3 and Gangapur Road's SO2 have low R^2 for both the model and persistence: these series are dominated by short, isolated spikes that no next-hour method anticipates.

12. Baseline Comparison

The persistence baseline predicts each pollutant at hour t+1 as its most recent observed value at or before hour t. It is evaluated on exactly the same station, hour and pollutant rows as the CNN-LSTM.

*Table 12. CNN-LSTM against persistence on identical test hours (original units). * = quality-flagged series.*

Station	Pollutant	n	RMSE CNN-LSTM	RMSE persist.	MAE CNN-LSTM	MAE persist.	R^2 CNN-LSTM	R^2 persist.	RMSE skill
MIDC Ambad	PM2.5	4542	5.976	6.106	3.981	3.954	0.8656	0.8597	0.02138
MIDC Ambad	PM10	4544	15.45	17.98	8.786	9.564	0.8544	0.8027	0.1407
MIDC Ambad	NO2	4467	5.086	5.958	2.794	3.139	0.8402	0.7807	0.1463
MIDC Ambad	SO2	4542	2.393	2.496	1.278	1.18	0.8637	0.8516	0.04153
MIDC Ambad	CO *	1574	0.08449	0.08424	0.05104	0.04811	0.4433	0.4466	-0.002942
MIDC Ambad	O3	4552	6.446	7.68	4.464	5.207	0.8903	0.8443	0.1606
MIDC Ambad	NH3	4489	1.381	1.378	0.8995	0.8446	0.7476	0.7489	-0.0026
Gangapur Road	PM2.5	4800	11.64	12.26	5.337	5.598	0.7073	0.6753	0.05065
Gangapur Road	PM10	4780	18.71	20.07	10.8	11.72	0.8108	0.7823	0.06767
Gangapur Road	NO2 *	2190	1.194	1.095	0.4078	0.08097	-1.101	-0.7663	-0.09071
Gangapur Road	SO2	4259	2.719	2.898	1.263	1.266	0.2832	0.1856	0.06186
Gangapur Road	CO *	1705	0.06184	0.0119	0.05053	0.007349	-0.5708	0.9418	-4.196
Gangapur Road	O3	4905	7.315	7.344	3.199	2.526	0.687	0.6845	0.003936
Gangapur Road	NH3 *	4198	1.978	2.038	0.3034	0.161	-0.8288	-0.9419	0.02957
Pandav Nagari	PM2.5	8135	9.652	10.26	3.762	3.95	0.8659	0.8484	0.05936
Pandav Nagari	PM10	8161	14.86	16.38	7.161	7.72	0.8919	0.8686	0.09281
Pandav Nagari	NO2	8199	3.945	4.29	2.081	2.129	0.7819	0.7421	0.08038
Pandav Nagari	SO2	8177	5.85	6.486	1.999	2.023	0.6176	0.5298	0.09812
Pandav Nagari	CO	8266	0.1383	0.1584	0.07446	0.0832	0.8104	0.7511	0.1274
Pandav Nagari	O3	8202	7.069	8.28	4.604	5.251	0.8857	0.8432	0.1462
Pandav	NH3	8216	4.692	4.915	2.044	1.787	0.6167	0.5795	0.04524

Nagari									
Hirwadi	PM2.5	4632	4.497	4.789	3.079	3.238	0.9311	0.9219	0.06101
Hirwadi	PM10	4628	17.48	18.45	7.233	7.993	0.765	0.7383	0.05241
Hirwadi	NO2	4402	3.414	3.489	2.091	1.944	0.7002	0.6869	0.02148
Hirwadi	SO2	4436	8.06	8.695	2.881	2.836	0.5709	0.5005	0.07307
Hirwadi	CO	2407	0.1742	0.1842	0.08154	0.06864	0.773	0.7462	0.05419
Hirwadi	O3	4642	12.63	13.68	6.087	7.027	0.7411	0.696	0.07715
Hirwadi	NH3	4508	8.047	8.327	3.561	3.516	0.2009	0.1443	0.03363

Table 13. Pooled comparison per pollutant (unflagged stations). The level-design column shows the rejected design for reference.

Pollutant	RMSE CNN-LSTM	RMSE persist.	RMSE level design	MAE CNN-LSTM	MAE persist.	R^2 CNN-LSTM	R^2 persist.	RMSE skill
PM2.5	8.674	9.155	10.24	4.006	4.16	0.8423	0.8243	0.05253
PM10	16.44	17.99	19.72	8.296	9.022	0.8512	0.8217	0.08646
NO2	4.154	4.612	5.173	2.27	2.345	0.8123	0.7686	0.09937
SO2	5.404	5.892	6.033	1.883	1.862	0.6538	0.5885	0.08279
CO	0.1471	0.1646	0.1827	0.07605	0.07992	0.8062	0.7575	0.1061
O3	8.475	9.386	10.23	4.575	5.012	0.8339	0.7963	0.09704
NH3	5.288	5.494	5.394	2.143	1.994	0.4613	0.4186	0.03744

On unflagged series the CNN-LSTM has lower RMSE than persistence in 23 of 24 station-pollutant pairs. Pooled RMSE skill ranges from 3.7% (NH3) to 10.6% (CO). The improvement is consistent but small. Persistence is a strong baseline at a one-hour horizon because consecutive hourly averages are highly correlated. On MAE the picture is mixed: the CNN-LSTM has a higher MAE than persistence in 9 of the 24 unflagged pairs. The network reduces large errors around rapid changes (which dominate RMSE) but adds small errors during calm periods, where repeating the last value is already almost exact. The level design, which predicts concentrations directly, was worse than persistence for 6 of 7 pollutants on pooled test RMSE. Model complexity alone did not produce better predictions; the gain came from framing the task as a correction to persistence.

13. Graphs and Interpretation

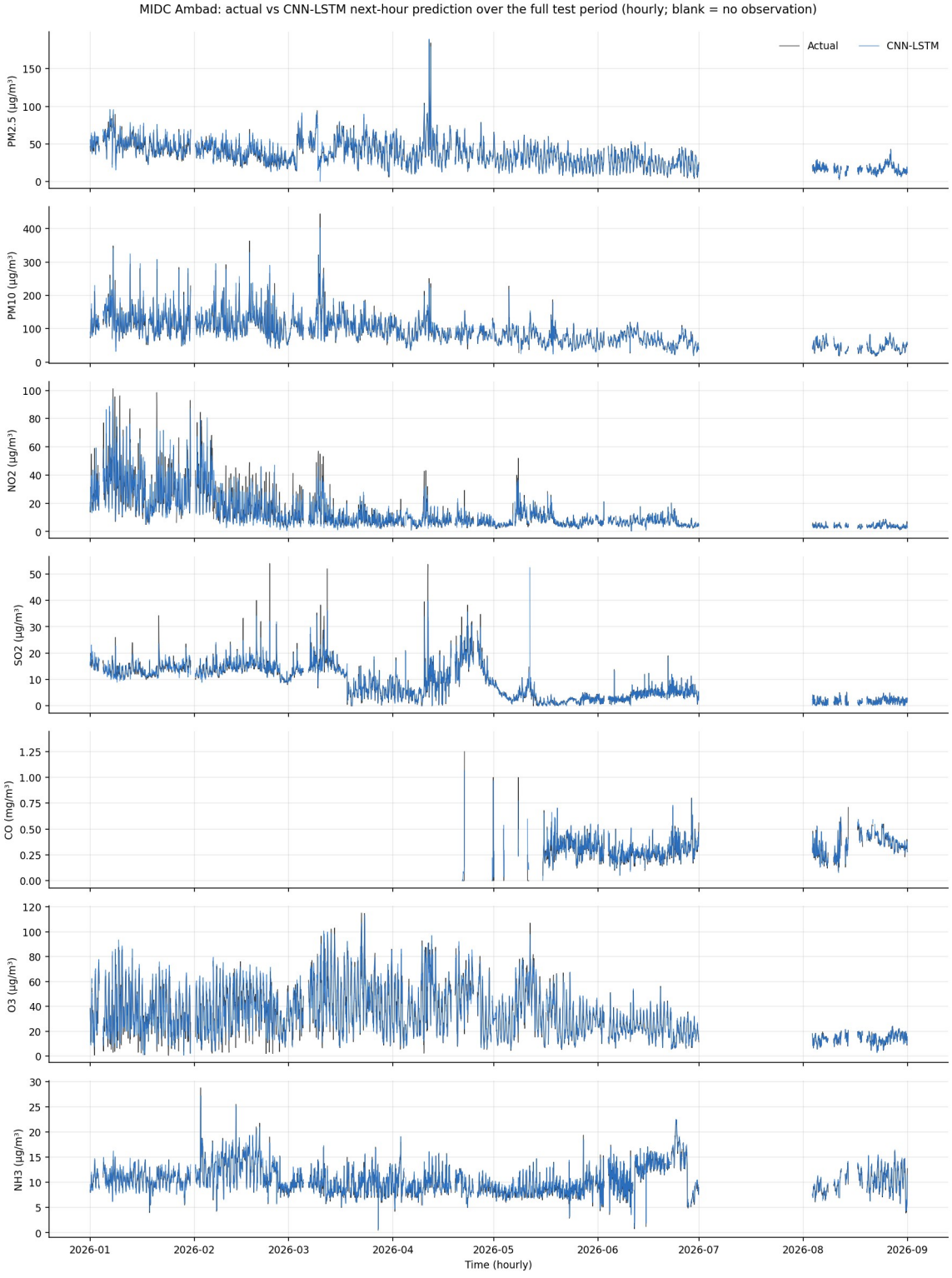


Figure 5. MIDC Ambad: hourly actual and CNN-LSTM predicted concentrations over the whole test period. Blank stretches have no observation (including July 2026).

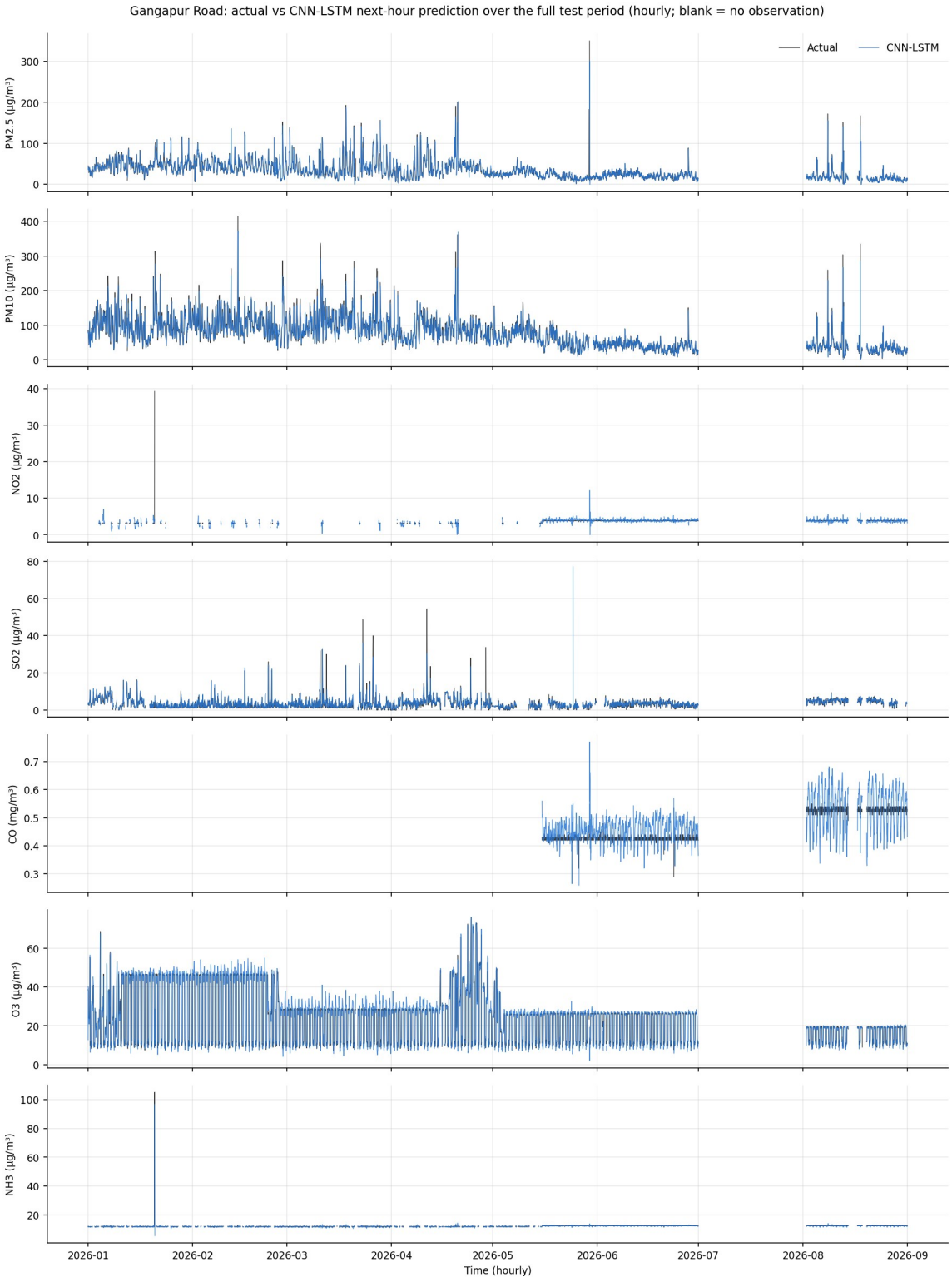


Figure 6. Gangapur Road: hourly actual and CNN-LSTM predicted concentrations over the whole test period. Blank stretches have no observation (including July 2026).

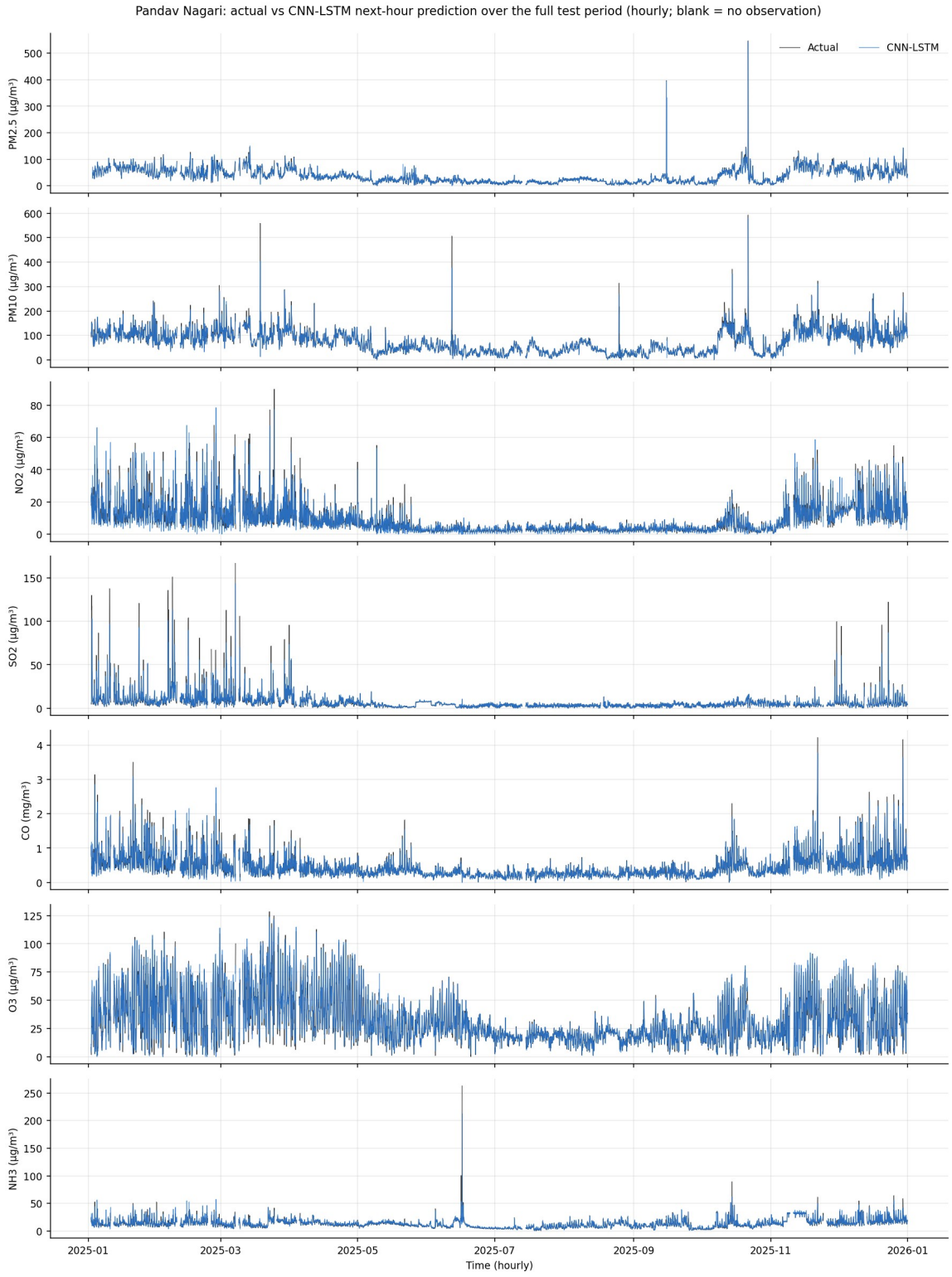


Figure 7. Pandav Nagari: hourly actual and CNN-LSTM predicted concentrations over the whole test period. Blank stretches have no observation (including July 2026).

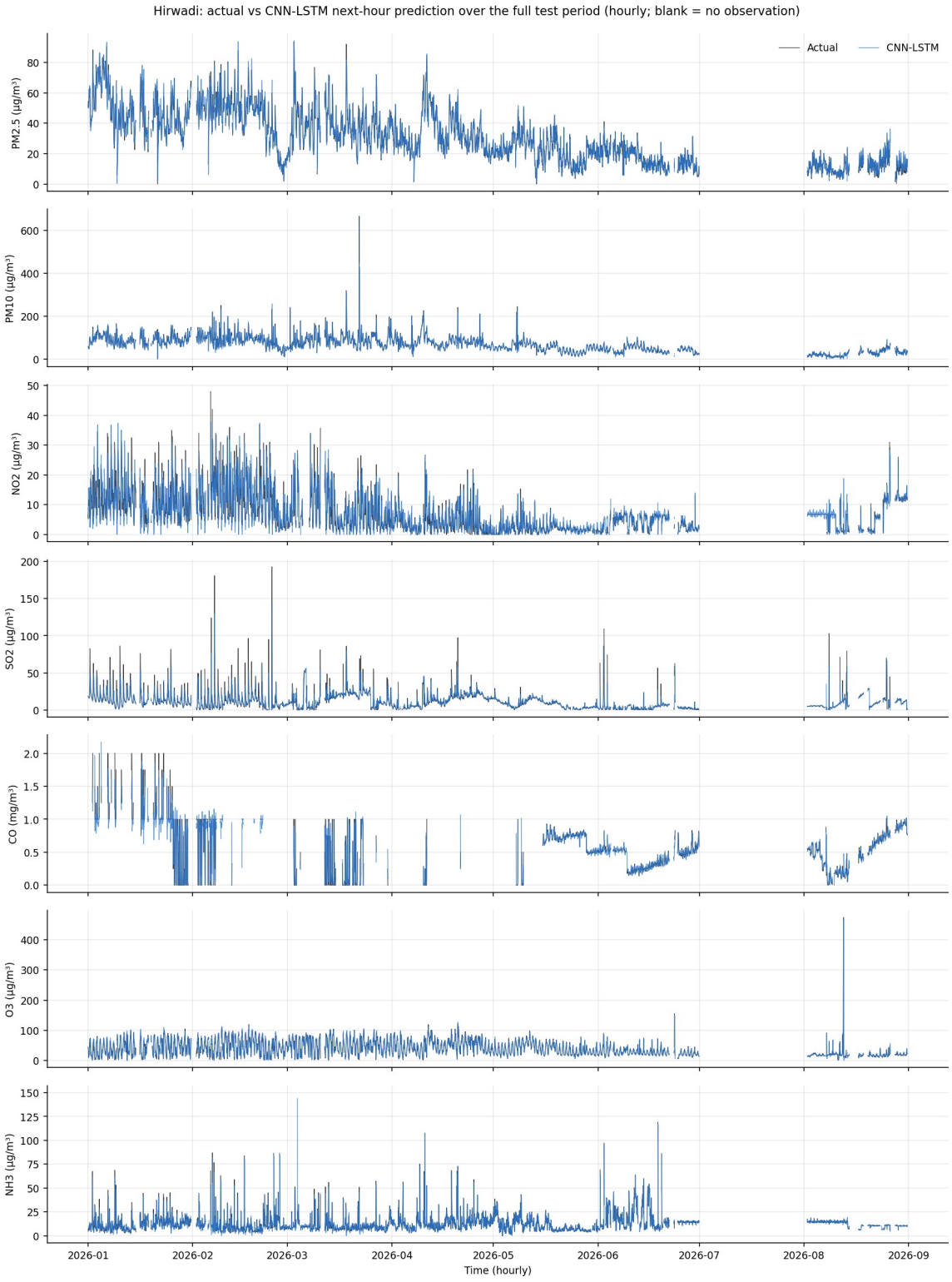


Figure 8. Hirwadi: hourly actual and CNN-LSTM predicted concentrations over the whole test period. Blank stretches have no observation (including July 2026).

Across the test period the predictions track the observed series closely at all stations, including the seasonal decline in particulate matter from winter to the monsoon. At hourly resolution the two lines overlap for most of the period, which is expected for a one-hour horizon. Visible departures are at isolated spikes (for example SO2 and NH3 bursts). For 28 of 28 series the predictions correlate more strongly with the previous hour's observation than with the current one: the model largely follows the series one hour late, as persistence does, and adds a small correction.

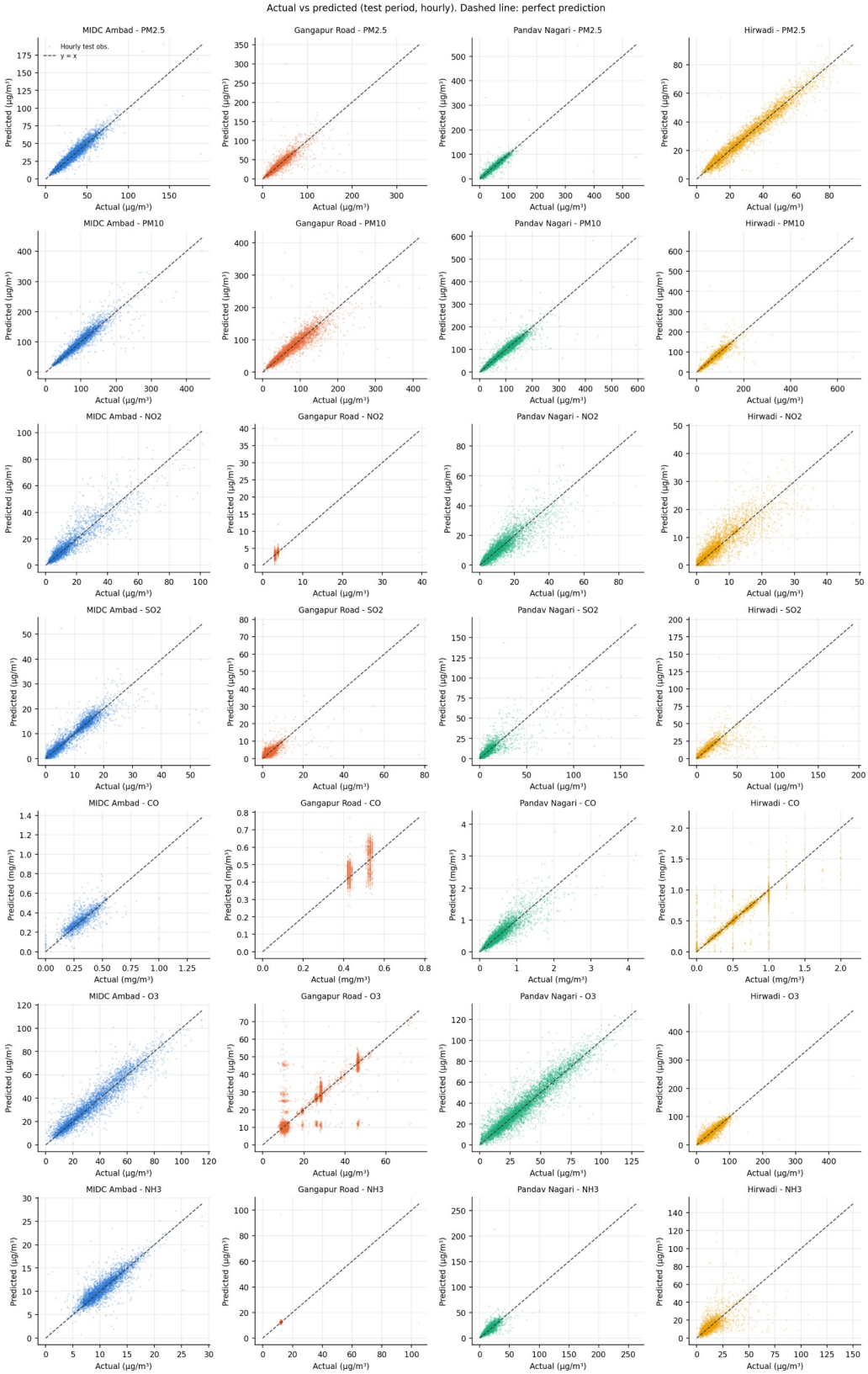

Actual vs predicted (test period, hourly). Dashed line: perfect prediction
MIDC Ambad - PM2.5
Hourly test obs.
y = x
Gangapur Road - PM2.5
Pandav Nagari - PM2.5
Hirwadi - PM2.5
MIDC Ambad - PM10
Gangapur Road - PM10
Pandav Nagari - PM10
Hirwadi - PM10
MIDC Ambad - NO2
Gangapur Road - NO2
Pandav Nagari - NO2
Hirwadi - NO2
MIDC Ambad - SO2
Gangapur Road - SO2
Pandav Nagari - SO2
Hirwadi - SO2
MIDC Ambad - CO
Gangapur Road - CO
Pandav Nagari - CO
Hirwadi - CO
MIDC Ambad - O3
Gangapur Road - O3
Pandav Nagari - O3
Hirwadi - O3
MIDC Ambad - NH3
Gangapur Road - NH3
Pandav Nagari - NH3
Hirwadi - NH3
Predicted (µg/m³)
Actual (µg/m³)
Predicted (mg/m³)
Actual (mg/m³)

Figure 9. Predicted against actual hourly concentration per station (columns) and pollutant (rows). Dashed line: perfect prediction.

Points cluster along the 1:1 line for particulate matter and O3. The flagged Gangapur Road CO panel shows vertical bands and its NO2 panel a tight cluster, because the observed values take only a few discrete levels. The highest values fall below the line: within the top 1% of observed hours of each series, the prediction is lower than the observation in a median of 82% of hours (range 35-98% across the 28 series).

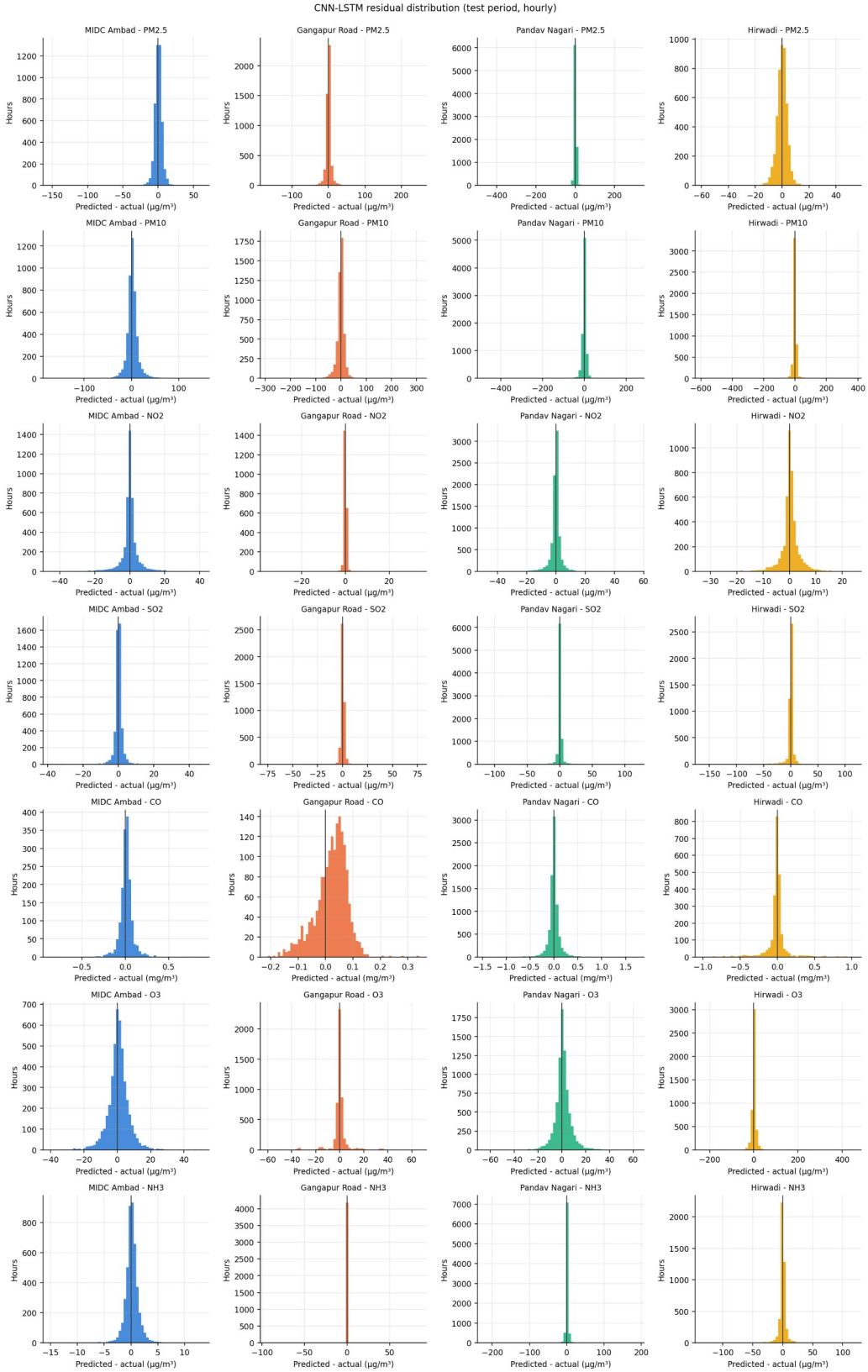


Figure 10. Distribution of CNN-LSTM residuals (predicted minus actual) per station and pollutant over the test period.

Residuals are centred near zero with long tails in both directions. The tails come from abrupt changes: in the hours with the sharpest rises, residuals are negative (the rise is under-predicted), and in the hours with the sharpest falls they are positive.

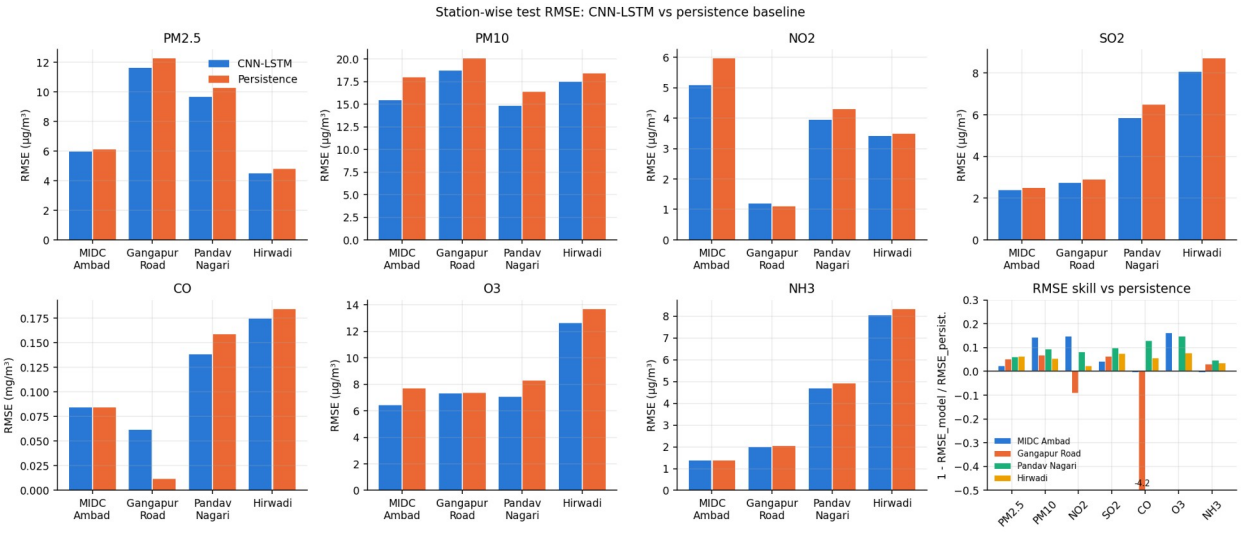


Figure 11. Test RMSE of the CNN-LSTM and persistence per station and pollutant, and RMSE skill (bottom right; bars below the axis limit are labelled with their value).

14. Discussion

Leakage checks. The scaler and imputer of each station were fitted on training rows only. A separate verification script recomputes the scaler means from the raw training rows, re-derives every test actual value and every persistence prediction from the raw CSV files, confirms that every evaluated hour lies in the test year, and recomputes MSE, RMSE, MAE and R^2 by hand from the exported predictions. The script was itself tested by injecting a leaked baseline, a one-hour timestamp shift and corrupted predictions; it rejected all three.

Quality-flagged series. Four test series take fewer than 100 distinct values over thousands of hours: Gangapur Road NO2, CO and NH3, and MIDC Ambad CO. At Gangapur Road in 2026, NO2 reads mostly 3.00 µg/m³ in 0.25 steps (34 distinct values, against about 2,400 per year before) and CO mostly 0.00 mg/m³ (17 values). The model was trained on the earlier, continuous behaviour, so its predictions are poor there (Gangapur Road CO skill -420%). These series measure an instrument change, not the model's ability to predict air quality. They are reported in every table but excluded from the pooled summary.

Overfitting and data availability. Early stopping ended training after 11-22 epochs. Hirwadi shows the clearest overfitting (validation loss rising after epoch 5). Training data is unbalanced: Gangapur Road has about 65,000 training windows, while Pandav Nagari has about 9,500. Despite that, Pandav Nagari has the best skill, which suggests that recent, consistent data matters more here than volume. Gangapur Road's long history includes years whose behaviour differs from 2026.

Comparison with other studies. Studies such as [3] and [4] report CNN-LSTM models outperforming other neural networks for PM2.5 prediction. The results here neither confirm nor contradict them. The experiment shows that, for next-hour prediction on these stations, a CNN-LSTM gives a small but consistent improvement over persistence when trained to predict the change from the last value, but not when trained to predict the level.

15. Limitations

- The latest year is incomplete: 2026 covers January-June and August (5,088 hours); July is absent. Results for 2026 do not represent a full seasonal cycle.
- Pandav Nagari is evaluated on 2025, not 2026, so station results cover different periods and are not strictly comparable.
- Three stations have only about two years of training data (mid-2023 to June 2025), and 2023 is partially covered.
- Meteorological inputs were excluded because of missing or inconsistent sensors. Wind, temperature and humidity are known drivers of pollutant concentration, and their absence limits accuracy, especially at spikes.
- One hour ahead only. Skill at longer horizons, where persistence degrades, was not measured.
- Single training run per model (seed 42); no confidence intervals or multiple-seed variance are reported. Differences of a few per cent in skill may be within run-to-run variation.
- The quality flag (fewer than 100 distinct test values) and the 12-hour stuck rule are simple heuristics; some degraded readings may remain (for example, Hirwadi CO alternates between about 0 and 1 mg/m^3 in February-April 2026).
- KNN imputation assumes hours with similar pollutant profiles are interchangeable. Long all-missing stretches receive the training mean.

16. Conclusion and Future Scope

A reproducible pipeline was built for next-hour prediction of seven pollutants at four Nashik stations, using leakage-free KNN imputation, per-station CNN-LSTM models and a chronological hold-out of each station's latest year. On quality-checked series, the selected model explains 0.46-0.85 of the pooled test variance and reduces RMSE relative to persistence by 3.7-10.6% depending on the pollutant. The improvement is modest, and a model that predicted levels directly did not beat persistence on validation data at any station. The experiment also exposed sensor problems in the 2026 data that should be resolved with the data provider.

Future work: add quality-controlled meteorological inputs; evaluate horizons of 6-24 hours, where persistence is weaker; report variance over several seeds; compare with simpler learned baselines (for example, gradient-boosted trees on lag features) to establish whether the recurrent architecture is necessary; and re-evaluate on a complete 2026 once the data is available.

17. References

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